Lead Scoring Case Study – Summary Report

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Objective

X Education converts only a minority of its leads, so sales effort is wasted on cold contacts. My task was to build a logistic regression model that gives every lead a score from 0 to 100, so the team can focus on hot leads, aiming for about 80% conversion among the leads worked.

How I approached it

Data cleaning. The 9,240-lead file had no duplicates, but the dropdown value "Select" is a hidden null, so I converted it to missing. I dropped seven columns with over 40% missing values and fourteen where one level covered over 95% of rows. Missing categories proved informative (blank occupation converts at 14% versus 49%), so I kept them as "Not Specified" instead of filling with the mode. I imputed numeric gaps with the median, capped outliers at the 99th percentile, merged levels under 1% into "Others", fixed duplicate spellings, and created one new metric, average time per visit.

EDA. Conversion rates by category showed that form and referral sources, working professionals, time on website and sales follow-up status separate buyers best.

Modelling. After creating dummy variables, I split the data 70:30, stratified, and standardised numeric columns using training data only. Rather than guessing how many features RFE should keep, I tuned it with five-fold cross-validation (25), then removed variables one at a time by p-value (above 0.05) and VIF (above 5), leaving 23. I chose the 0.35 cut-off from accuracy, sensitivity, specificity and precision-recall curves. I compared the result with a decision tree, a random forest, a smaller RFE model and a model without Tags.

Results

On unseen data the model reaches AUC 0.976, accuracy 93.5%, sensitivity 92.9% and specificity 93.9%, almost identical to training, so it is not overfitted. Leads scored hot (above 35) convert at 91% against a 38.5% base rate, beating the CEO's 80% target, and the top 30% of leads hold 76% of all conversions. Moving the cut-off answers the business questions: 0.15 for intern season (96% of buyers reached, 84% conversion of calls) and 0.80 once the quarterly target is met (95% of calls convert, useless calls cut by 57%).

Lessons learnt

1. Cleaning decisions matter as much as modelling: "Select" and missing values needed thought, not automatic imputation.
2. Honest evaluation means selection, scaling and cut-off choice use training data only.
3. Simpler is better: the interpretable logistic regression matched a random forest.
4. Beware very strong features: Tags is recorded by sales during follow-up. Without it, accuracy falls to about 80% (AUC 0.88), so I state this assumption and report both models.
5. A model matters only when tied to decisions, so I translated cut-offs into calling policies.